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Neuroadaptive integral robust control of visual quadrotor for tracking a moving object

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ABSTRACT

This paper aims to achieve the visual quadrotor tracking for ground moving target in the presence of various uncertainties. In order to provide a stable quadrotor motion, the perspective image moments extracted from the defined virtual image plane are selected as the visual features to deduce a decoupled visual quadrotor model. Then a neuroadaptive integral robust controller is designed for the outer and inner loops of visual quadrotor within the image based visual servo (IBVS) framework, which enables error stabilization and improves the anti-disturbance capability. Then, the unknown lumped uncertainties consisting of parametric uncertainties and external disturbances are estimated via constructing a minimal learning parameter-neural network (MLP-NN) and compensated in the feedforward loop, subsequently the residual estimation errors are further suppressed by the structured robust integral of the sign of the error (RISE) feedback control law, hence a high-precise servo tracking performance can be expected. The innovations of the algorithm lie in, (1) with the NN approximation compensated in feedforward loop, the residual estimation errors in feedback loop are decreased, hence the conservatism that using high gains when facing large uncertainties is relaxed, (2) by means of the peculiarity of RISE control, asymptotic stability can be guaranteed with continuous and bounded control inputs, (3) using MLP technology, the number of NN learning parameters are decreased to only two for the closed-loop system, and the online computational load is greatly reduced. Finally, the validity and superiority of the proposed method are validated through affluent simulations and comparisons.

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1. Introduction

Quadrotor unmanned aerial vehicles (UAVs) have been extensively studied in recent years and received much attention in both military and civilian scenarios due to mechanical simplicity, low development/maintenance cost and high hovering precision, etc [1,2]. Many significant applications are achieved using quadrotor platform, such as motion capture [3], search and localization [4], payload delivery [5], to name just a few. Notably, autonomous target tracking is an essential capability to enforce these missions [6,7]. For the majority of researches on target tracking of quadrotor, vision sensor is typically equipped and regarded as a primary unit to localize the target [8]. Moreover, it can also provide the translational velocity signal of the quadrotor in combination with the inertial measurement unit (IMU), independent of global position (GPS) system.

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The control strategy using the visual information to control the quadrotor motion relative to a target is called as visual servo (VS) [9,10], which can be classified into two main frameworks: position based visual servo (PBVS) and image based visual servo (IBVS). The PBVS, using image information to reconstruct the three dimensional (3D) position, heavily relies on the prior target geometric model and precise camera calibration [11]. In contrast, for the latter, IBVS calculates the control law only through 2D image errors [12]. Hence, the IBVS has a more clear and robust control structure with low computational complexity. With the above advantages, the IBVS has gradually become the mainstream in the visual servo field and has been widely used for manipulator [13] and robot [14].

For the IBVS control design, the errors defined in the image plane are applied to regulate the quadrotor behaviors, hence the selection of image features is vital to enable the quadrotor motion stabilization [12]. The existing features, such as the point, the line, and the ellipse features, are generally subject to the image singularities caused by the environmental change, like light condition [15]. To solve the singularity problem, image moments are proposed in [16] for the visual servo control. The moment features give a universal description of any object, with simple or complex shapes, which can be computed from the segmented image, and have a geometric and intuitive significance, which can be attributed to the heavy dependence on the centroid, the area, and the orientation of the object in the image. However, the spherical image moments used in the above literature cannot provide the proper behavior for quadrotor in vertical axis as experimentally displayed in [17]. To overcome this issue, perspective image moments are introduced in [18] to deduce a dynamic IBVS approach. Also, a virtual image plane is employed to project the perspective image moments, where the passivity-like properties hold and the decoupled image dynamics can be obtained. Using these technologies, an appropriate 3D motions of quadrotor in the workspace can be provided, and the controller design is greatly simplified. As well, the idea has also been extended and experimentally verified by researchers in [19,20].

Based on decoupled features, the desired IBVS control approach can yield appropriate velocity commands in 3D space, further achieve satisfactory changes in the image. However, it still faces some difficulties to design an excellent servo controller for quadrotor due to the strong nonlinearity nature, together with unknown uncertainties including parametric uncertainties and external disturbances. In [21,22], back-stepping control is utilized as a fundamental framework to design the IBVS controller, which treats each differential equation in the system as a virtual system, and deduces the virtual control law by the recursive utilization of Lyapunov functions. However, this control approach lacks performance robustness against uncertainties. To enhance the robustness, the literature [23] treats the uncertainties as unknown system parameters, and develops an adaptive controller to deal with the uncertainties and reduce the steady-state errors. Nevertheless, only the constant disturbances including aerodynamics constants and attitude estimation bias are considered in the abovementioned paper, which is too ideal to test the performance of controller. When facing vast range of external disturbances, the adaptive control may be unstable. Moreover, the observer based control, i.e., extended state observer (ESO), is also employed in quadrotor system for suppressing the uncertainties [24,25]. However, this method only can guarantee the semiglobally uniformly ultimate boundedness (UUB) of the controlled system due to the inability in eliminating the residual estimation errors.

Recently, a continuous robust integral of the sign of the error (RISE) control is presented in [26], with a model based feedforward control term in nature to deal with the nonlinear behaviors of the system and a particular robustified nonlinear control term to compensate for sufficiently smooth bounded disturbances. The most attractive characteristic of this control approach is that asymptotic stability can be achieved for uncertain nonlinear systems subject to modeling uncertainties, which has been widely used in reality for DC motor [27] and autonomous underwater vehicle [28]. Unfortunately, high feedback gains are still required when facing large uncertainties, which is generally undesirable in practice. In order to provide a satisfactory control performance, substantial improvements have been made to reduce the feedback gains of RISE control. In [29], by a fusion of RISE control, adaptive control and proportional integral control, an accurate model reference tracking can be achieved with weak feedback gains. In [30], an ESO in combination with RISE control is presented for suppressing the model parametric uncertainties and external disturbances, where the uncertainty estimation via ESO is compensated in feedforward loop to reduce the control efforts. Especially, by means of the universal approximation capability of neural networks (NNs), the NN approximator based adaptive control schemes are often incorporated with the RISE control [31,32] to reject the effect of various uncertainties and provide a precise control performance. As well, the feedback gains in the robustified nonlinear control term can be further decreased. However, the aforementioned NN control approaches generally suffer from the explosion of learning parameters. For the NN approximator, the number of neural weights in hidden units that requires to be tuned online, will significantly grow with the dimension of the parameter vector of the unknown functions [33], indicating that increased design complexity and tedious parameter tuning are inevitable. And the excessive adaptive computational burden will further lead to certain time delays in controller computing and actuation, sometimes may cause the loss of target during the visual tracking for complicated targeting route. Fortunately, by replacing the ideal weight vectors with their maximum norm, the minimal learning parameter (MLP) technology is proposed in [34] to reduce the computational load of NN weights, such that the dimension of learning parameters can be decreased without sacrificing the identification accuracy.

Inspired by the above works, in this study, we put forward a neuroadaptive integral robust controller for the visual quadrotor to enforce the moving object tracking mission in the presence of parametric uncertainties, external disturbances and measurement noises. A virtual camera is first introduced to project the defined perspective image moments. With that, a decoupled visual quadrotor dynamics is derived, which isolates the roll and pitch motions and facilitates the control design. Then a neuroadaptive integral robust based IBVS control is presented for quadrotors to stabilize the system and suppress the

internal and external uncertainties, where a neuroadaptive control law with MLP is designed to approximate the unknown uncertainty with only two learning parameters for the total system, and the residual approximation errors are further restrained by the RISE feedback control law. Hence, a more precise servo control performance can be provided. The main contributions of this paper lie in:

- A neuroadaptive integral robust based IBVS scheme with MLP technology is proposed for the visual quadrotor to carry out the ground target tracking with complicated routes in the presence of various uncertainties. The designed controller can guarantee accurate 3D tracking with asymptotic tracking performance and strong robustness against uncertainties.
- By comparison with the traditional RISE controllers [26–28] that use a robustified nonlinear control term to restrain all internal and external uncertainties, in our work, the lumped uncertainties are first estimated by the designed neural approximator and compensated in feedforward loop, hence the RISE feedback control law only needs to suppress the residual NN estimation errors. Thus, the design conservatism on selecting appropriate gains for RISE control term is relaxed.
- In contrast to traditional NN designs [32,35] that only realize uniformly ultimate boundedness due to the unavoidable estimation errors of neural approximators, by incorporating the RISE control into adaptive NN, the present method can ensure the asymptotic stability of system with guaranteed tracking precise in the image and 3D space. This property is also theoretically proved by the Lyapunov analysis, where all the closed-loop tracking errors under the present neuroadaptive integral robust based IBVS scheme converge to zero ultimately.
- By means of the MLP technology, only two learning parameters are required for neural approximation in the total system, thus the adaptive computational burden is greatly reduced. Compared to the general NN designs [31,33] that require a large number of online tuning arguments, the designed controller is more feasible in reality with low computational complexity.

In Section 2, the nonlinear dynamics model of visual quadrotor system is introduced and preliminaries are described briefly. The Neuroadaptive integral robust based IBVS controller for quadrotor is given in Section 3. Then the stability of controller is analyzed in Section 4. Plentiful simulations and comparisons are illustrated in Section 5. Finally, Section 6 draws some conclusions.

2. Visual quadrotor model and preliminaries

2.1. Visual quadrotor model

To deduce the image and quadrotor dynamics, the relative reference frames and image planes are illustrated in Fig. 1. Two coordinate systems are first defined: the inertia frame $\Phi_I = \{O_i, X_i, Y_i, Z_i\}$, and the body frame $\Phi_B = \{O_b, X_b, Y_b, Z_b\}$ locating at the center of mass of the quadrotor. Then the camera frame $\Phi_C = \{O_c, X_c, Y_c, Z_c\}$ is established to derive the image dynamics, which is assumed to align with the quadrotor body frame Φ_B. The image plane related to camera frame is defined in a standard distance (f) from the quadrotor with f being the camera focal length, and parallel to the $X_c - Y_c$ plane. Next the virtual frame $\Phi_V = \{O_v, X_v, Y_v, Z_v\}$ is given to deduce the decoupled image feature dynamics, which is produced by rotating the camera frame Φ_C such that Φ_V has zero roll and pitch angles and the position and yaw angle of Φ_V are equivalent with Φ_C [19]. Owing to the use of the virtual frame, the image kinematics is independent of roll and pitch rates, which is

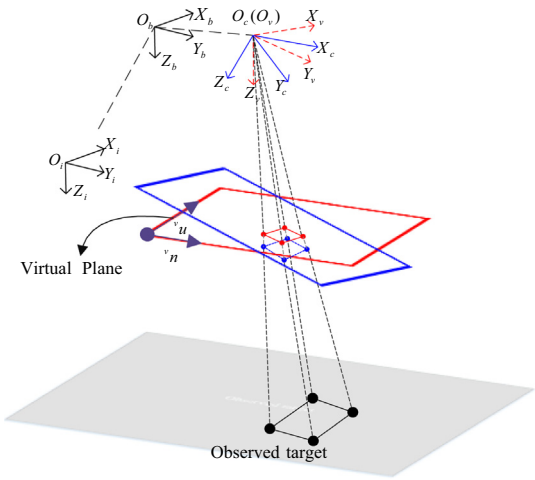

Fig. 1. The coordinate frames and image planes.

significant in simplifying the image kinematics and facilitating the control design. Also, a virtual image plane related to the virtual camera is defined, where their alignment is the same to those of camera framework and real image plane.

For the IBVS controller design, a ground object with N non-collinear points is generally selected as the target [18–20]. Its coordinates in the virtual image plane are assumed as $({}^{v}u_k, {}^{v}n_k), k = 1, 2 \ldots N$. Then the decoupled low order perspective moments related to the centroid, the area, and the orientation of the object are chosen as image features to deduce the independent image dynamics:

$$q_x = q_z \frac{{}^{v}u_g}{f}, q_y = q_z \frac{{}^{v}n_g}{f}, q_z = \sqrt{\frac{a^*}{a}}, \alpha = \frac{1}{2}\arctan\left(\frac{2\,{}^{v}\mu_{11}}{{}^{v}\mu_{20} - {}^{v}\mu_{02}}\right) \tag{1}$$

In above, $a = {}^{v}\mu_{20} + {}^{v}\mu_{02}$ is the area of the object in the virtual image plane with ${}^{v}\mu_{ij} = \sum_{k=1}^{N}({}^{v}u_k - {}^{v}u_g)^i({}^{v}n_k - {}^{v}n_g)^j$ being the central moment of an object's image, ${}^{v}u_g = {}^{v}m_{10}/{}^{v}m_{00}, {}^{v}n_g = {}^{v}m_{01}/{}^{v}m_{00}$ can be regarded as the center of gravity of an object in the image, where ${}^{v}m_{ij} = \sum_{k=1}^{N} {}^{v}u_k^i\, {}^{v}n_k^j$ is the image moment. And, a^* is the desired value of a and satisfies $z\sqrt{a} = z^*\sqrt{a^*}$, where z denotes the distance from the quadrotor to the considered target and z^* is its desired value. For the selected image features, q_z is inversely proportional to a, hence we can conclude that the changes of such features depend linearly on the depth. Furthermore, the design of q_x and q_y allows us to completely partition the three selected features to the three translational DOFs. This feature can facilitate the subsequent IBVS controller design.

With the dynamics of image point in image plane introduced by [36], the dynamics of image point in the virtual image plane can be deduced with the transformation among the above frameworks:

$$\begin{bmatrix} {}^{v}\dot{u} \\ {}^{v}\dot{n} \end{bmatrix} = \begin{bmatrix} -\frac{f}{{}^{v}z} & 0 & \frac{{}^{v}u}{{}^{v}z} \\ 0 & -\frac{f}{{}^{v}z} & \frac{{}^{v}n}{{}^{v}z} \end{bmatrix} \begin{bmatrix} vv_x \\ vv_y \\ vv_z \end{bmatrix} + \begin{bmatrix} {}^{v}n \\ -vu \end{bmatrix} \dot{\psi} \tag{2}$$

where $({}^{v}x, vy, vz)$ are the coordinates of object point expressed in virtual frame, $({}^{v}u, {}^{v}n)$ indicate the coordinates of projecting point in the virtual image plane and ψ is the yaw angle of camera.

With the help of , we can further derive the dynamics of the defined image moments in the virtual image plane:

$$\dot{\mathbf{q}} = -sk(\dot{\psi}\boldsymbol{\delta}_{03})\mathbf{q} - v\mathbf{V}/z^*, \dot{\alpha} = -\dot{\psi} \tag{3}$$

where $\mathbf{q} = [q_x, q_y, q_z]^T$ is the defined image feature vector, ${}^{v}\mathbf{V}$ is the linear velocity of camera framework with respect to the virtual frame. $\boldsymbol{\delta}_{03} = [0, 0, 1]^T$ means the vector of unit length in the inertia frame. And expression $sk(\cdot)$ is a skew-symmetric matrix.

Recalling the quadrotor dynamics in [22,24], the translational and rotational motion equations of the vision-quadrotor system can be given by:

$$\begin{cases} \dot{\mathbf{q}} = -sk(\dot{\psi}\boldsymbol{\delta}_{03})\mathbf{q} - \mathbf{R}_{\psi}^{T}(t)\,{}^{i}\mathbf{V}/z^* \\ m\,{}^{i}\dot{\mathbf{V}} = -\boldsymbol{\Pi}_1^{i}\mathbf{V} - \mathbf{G} + \mathbf{g}_1(\boldsymbol{\Theta}, \psi)U_1 + \mathbf{d}_v(\mathrm{t}) \\ \dot{\boldsymbol{\Theta}} = \boldsymbol{\delta}_{23}\boldsymbol{\gamma} \\ \mathbf{J}\dot{\boldsymbol{\gamma}} = -\boldsymbol{\Pi}_2\boldsymbol{\gamma} + \mathbf{g}_2\boldsymbol{\Gamma} + \mathbf{d}_{\gamma}(\mathrm{t}) \end{cases} \tag{4}$$

where $\mathbf{R}_{\psi}^{T}(t)$ denotes the rotation matrix around Z_i, reflecting the transforming relationship between virtual framework and the inertial framework. ${}^{i}\mathbf{V}$ is linear velocity of camera framework related to the inertial framework. $\boldsymbol{\Theta} = [\phi, \theta]^T$ denotes the quadrotor orientation, with ϕ, θ being the roll, pitch angles respectively, and ψ expresses the yaw angle. $\boldsymbol{\gamma} = \left[\gamma_{\phi}, \gamma_{\theta}, \gamma_{\psi}\right]^T$ is the angular rate of the quadrotor in body frame. $\boldsymbol{\delta}_{23} = [\boldsymbol{\delta}_{01}, \boldsymbol{\delta}_{02}]^T$ with $\boldsymbol{\delta}_{01} = [1, 0, 0]^T$ and $\boldsymbol{\delta}_{01} = [1, 0, 0]^T$. $m \in \mathbb{R}^+$ refers to the mass of quadrotor and $\mathbf{G} = [0, 0, mg]^T$ with g being the gravity acceleration. $\mathbf{J} = diag(J_1, J_2, J_3) \in \mathbb{R}^{3\times3}$ is a positive definite diagonal inertial matrix with $J_i (i = 1..3)$ being the inertial moment. $\boldsymbol{\Pi}_1 = diag(k_x, k_y, k_z)$,$\boldsymbol{\Pi}_2 = diag(k_{\phi}, k_{\theta}, k_{\psi})$ denote damping matrices, where $k_i (i = x, y \ldots, \psi)$ are drag coefficients. $\mathbf{g}_1(\boldsymbol{\Theta}, \psi) = [c(\psi)s(\theta)c(\phi) + s(\psi)s(\phi), s(\psi)s(\theta)c(\phi) - c(\psi)s(\phi), c(\theta)c(\phi)]^T$ is the input matrix of position subsystem correlated with the orientation, in which $s(x), c(x)$ are short for $sin(x)$ and $cos(x)$. $\mathbf{g}_2 = diag(l, l, c) \in \mathbb{R}^{3\times3}$ with l, c respectively being the distance from the quadrotor's mass center to each rotor and a constant force-to-moment factor. The control input signals $U_1 \in \mathbb{R}$ and $\boldsymbol{\Gamma} = [U_2, U_3, U_4]^T$, represent the applied thrust force and three rotational torques elements along $X_i - Y_i - Z_i$ axis, respectively. $\mathbf{d}_v(t) = \left[d_x, d_y, d_z\right]^T, \mathbf{d}_{\gamma}(t) = \left[d_{\phi}, d_{\theta}, d_{\psi}\right]^T$ are the bounded external disturbances existing in position and attitude dynamics.

The yaw rotational dynamics of quadrotor can be obtained using the object orientation α defined by (1), and its time derivative is

$$\dot{\alpha} = -\gamma_{\psi} \tag{5}$$

For the sake of simplicity, we define the following notations:

$$\begin{aligned}
&\mathbf{X}_1 = \mathbf{q}, \mathbf{X}_2 = -sk(\dot{\psi}\boldsymbol{\delta}_{03})\mathbf{q} - \mathbf{R}_{\psi}^T(t)\,{}^i\mathbf{V}/z^*, \mathbf{X}_3 = [\phi, \theta, \alpha]^T, \mathbf{X}_4 = \left[\gamma_\phi, \gamma_\theta, \gamma_\psi\right]^T, \\
&\mathbf{F}(U_1, \boldsymbol{\Theta}, \psi) = -(\mathbf{g_1}U_1 - \mathbf{G})/m, \boldsymbol{\Pi}_1 = \bar{\boldsymbol{\Pi}}_1 + \Delta\boldsymbol{\Pi}_1, \mathbf{f_V} = \mu_1\mathbf{x}_2 + \mu_2\mathbf{x}_1 \\
&\mu_1 = -z^*(sk(\dot{\psi}\boldsymbol{\delta}_{03}) + \mathbf{R}_{\psi}^T(t)\boldsymbol{\Pi}_1), \mu_2 = -z^*(sk(\ddot{\psi}\boldsymbol{\delta}_{03}) + \mathbf{R}_{\psi}^T(t)\boldsymbol{\Pi}_1 sk(\dot{\psi}\boldsymbol{\delta}_{03})), \mathbf{d}_1 = \left(\Delta\boldsymbol{\Pi}_1^i\mathbf{V} - \mathbf{d}_v(t)\right)/m \\
&\boldsymbol{\Pi}_2 = \bar{\boldsymbol{\Pi}}_2 + \triangle\boldsymbol{\Pi}_2, \mathbf{f}_\gamma = -\mathbf{J}^{-1}\bar{\boldsymbol{\Pi}}_2\boldsymbol{\gamma}, \mathbf{g}_2 = \bar{\mathbf{g}}_2 + \Delta\mathbf{g}_2, \bar{\boldsymbol{\Gamma}} = \mathbf{J}^{-1}\bar{\mathbf{g}}_2\boldsymbol{\Gamma}, \mathbf{d}_2 = \mathbf{J}^{-1}\left(\mathbf{d}_\gamma(t) - \Delta\boldsymbol{\Pi}_2\boldsymbol{\gamma} + \Delta\mathbf{g}_2\boldsymbol{\Gamma}\right)
\end{aligned} \tag{6}$$

where $\mathbf{X_2}$ can be regarded as the optic flow of image features in the virtual image plane, and the linear velocity ${}^i\mathbf{V}$ can be computed by measuring $\mathbf{X_2}$ and offsetting the rotational term $sk(\dot{\psi}\boldsymbol{\delta}_{03})$. $\mathbf{f_V}, \mathbf{f}_\gamma$ are the nonlinear model terms in the dynamics of quadrotor. $\mathbf{F} \in \mathbb{R}^{3\times 1}$ denotes the virtual control input vector for quadrotor position dynamics. $\bar{\boldsymbol{\Gamma}}$ is the control vector for attitude loop. Δ denotes the parameterized uncertainties induced by the perturbation of aerodynamics factors and the superscript bar represents its nominal value. $\mathbf{d}_1$, $\mathbf{d}_2$ are unknown and can be regarded as the lumped uncertainties containing external disturbances and parameterized uncertainties existing in the system dynamics. Then the system dynamics of the vision-quadrotor can be converted into the following state-space form:

$$\begin{cases}
\dot{\mathbf{X}}_\mathbf{1} = \mathbf{X_2} \\
z^*\dot{\mathbf{X}}_\mathbf{2} = \mathbf{f_V} + \mathbf{F} + \mathbf{d_1} \\
\dot{\mathbf{X}}_3 = \boldsymbol{\rho}\mathbf{X}_4 \\
\dot{\mathbf{X}}_4 = \mathbf{f}_\gamma + \bar{\boldsymbol{\Gamma}} + \mathbf{d_2}
\end{cases} \tag{7}$$

with $\boldsymbol{\rho} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

Remark 1. In contrast to the existing works, the control issue considered is more complicated. First, unlike the model formulation of a quadrotor explored by [23] that only considers aerodynamics constants and attitude estimation bias, the uncertainties consisting of parameterized uncertainties, time-varying external disturbances are synchronously taken into account, which is more practical to prove the robustness performance of the control methodology against uncertainness. Second, the visual quadrotor dynamics is constructed in inertia frame, which is more straightforward to reflex the quadrotor motions by comparison with the previous visual servo strategies [18–20].

Remark 2. In comparison with visual servo algorithms using spherical image moments as the image features [16], which cannot universally provide appropriate behavior for quadrotors in vertical axis, the selected perspective image moments can achieve stable translational and yaw rotational motions in 3D space with the introduced virtual plane. As seen from (7) that a decoupled and linear relationship between the object image dynamics and the quadrotor motion is given by exploiting the selected moment features in the defined virtual image plane, and also the passivity properties of feature dynamics mentioned in [18] are preserved.

Assumption1 (*[36]*). The unknown lumped uncertainties $\mathbf{d}_1, \mathbf{d}_2$ and their derivatives are bounded and satisfy $\| \mathbf{d}_1 \| \leqslant D_1, \| \dot{\mathbf{d}}_1 \| \leqslant D_2, \| \mathbf{d}_2 \| \leqslant D_3, \| \dot{\mathbf{d}}_2 \| \leqslant D_4$ with D_1, D_2, D_3, D_4 being unknown constants.

Remark 3. The sources of uncertainties $\mathbf{d}_1, \mathbf{d}_2$ include wind disturbance, force-to-moment factor uncertainties, measurement errors of body arms, et al. The energy of these uncertainties is always finite in reality, hence it is reasonable to assume $\mathbf{d}_1, \mathbf{d}_2$ bounded by an unknown constant to facilitate the preceding stability analysis.

The objective of this work is to construct a neuroadaptive integral robust control based IBVS control scheme for the quadrotor to pursuit the ground target with complicated moving trajectories in the presence of various uncertainties. To this end, a camera mounted under the quadrotor is required to offer visual information. As well, velocities, Euler angles and angular rate information of the quadrotor are available for the IBVS control design via the optical flow arithmetic and IMU. The observed target is assumed as a planar object, whose initial pose is assumed adjacent to the quadrotor such that the object points maintain inside the camera's FOV initially.

2.2. MlP-Nn

Radial basis function neural network (RBFNN) is introduced here to learn the unknown nonlinearities and uncertainties by virtue of its excellent performance and global approximation ability. The MLP technology is further employed to reduce the number of NNs and tuning parameters. The kernel of RBFNN is given by the following lemma.

Lemma 1 ([34,37]). For any given continuous bounded function $d(\mathbf{X})$ with $d(0) = 0$, by virtue of the continuous function separation technique and RBFNN approximation technique, we have the following expression:

$$d(\mathbf{X}) = \mathbf{W}^{*T}\mathbf{h}(\mathbf{X}) + \varsigma \tag{8}$$

where ς represents the approximation error with unknown upper bound ς_M. $\mathbf{X} = [x_1, x_2, \ldots x_n]^T \in \mathbb{R}^n$ denotes the input vector of neural network, $\mathbf{W}^* = \left[w_1^*, w_2^*, \ldots w_p^*\right]^T \in \mathbb{R}^p$ is the weight vector. p and n respectively denote the node number and input number. $\mathbf{h}(\mathbf{X}) = [h_1(\mathbf{X}), h_2(\mathbf{X}), \ldots h_P(\mathbf{X})]^T \in \mathbb{R}^p$ and $h_j(\mathbf{X})$ is typically selected as

$$h_j(\mathbf{X}) = \exp\left(-\frac{\| \mathbf{x} - \mathbf{c} \|^2}{2b_j^2}\right), j = 1, 2, \ldots p \tag{9}$$

with $\mathbf{c} = [c_1, c_2, \ldots c_n]^T \in \mathbb{R}^n$ being the center vector and b_j being the variance representing the spread of the basis function.

With the help of mapping relationship in (8), we can employ the output of RBFNN to approximate the unknown function $d(\mathbf{X})$ with an approximation error ς. Notably, the desired weight vector is usually unknown. For the RBFNN, an adaptive update of weights is needed in practice. To reduce the computation, the MLP is used in this paper. This method defines a novel adapted weight $\varphi = \| \mathbf{W}^* \|^2$ to replace the weight vector $\mathbf{W}^*$. The parameters adjusted online for the NN are reduced to a great extent.

Remark 4. For the basis function (9), $\exp(\cdot)$ is a is a strictly monotonic increasing function and $-\| \mathbf{x} - \mathbf{c} \|^2/2b_j^2 \leqslant 0$, hence we can obtain that $0 < h_j(\mathbf{X}) \leqslant h_j(0) = 1$. So there exists a positive constant $\hbar$ satisfying $\| \mathbf{h}(\mathbf{X}) \| \leqslant \hbar$.

3. IBVS controller design

The control objective in the paper is to regulate the quadrotor motion to timely track the moving target in ground subject to various uncertainties, via matching the image features of the object with predefined moment features. To achieve this objective, a neuroadaptive integral robust controller is constructed in the section. As revealed by the graphical description in Fig. 2, the controller is divided into two loops: the outer (visual servo) loop with slow dynamics that drives the quadrotor to track a moving target using captured image information by the camera and the inner (attitude) loop with fast dynamics that accounts for regulating the orientation to timely track the desired attitude produced by the outer loop. For each loop, a neuroadaptive integral robust is designed for suppressing the unknown lumped uncertainties, where a neural approximator is used to identify the uncertainties, the estimated results and model nonlinearities are synchronously compensated by the feedforward control law, then the residual estimation errors are further eliminated via the designed robust feedback control law. Particularly, in order to address the explosion of learning parameters existing in the NN control, the MLP technology is utilized for the NN design with a novel adaptive updating law of weight. Using that, the number of learning parameters are reduced to only two for the entire system. In addition, by implanting the MLP-NN into the RISE control law, the design conservatism in choosing an appropriate feedback gain for integral signum term is greatly reduced and some rigorous prerequisites in establishing the stability analysis of RISE can be released, as demonstrated in the subsequent sections.

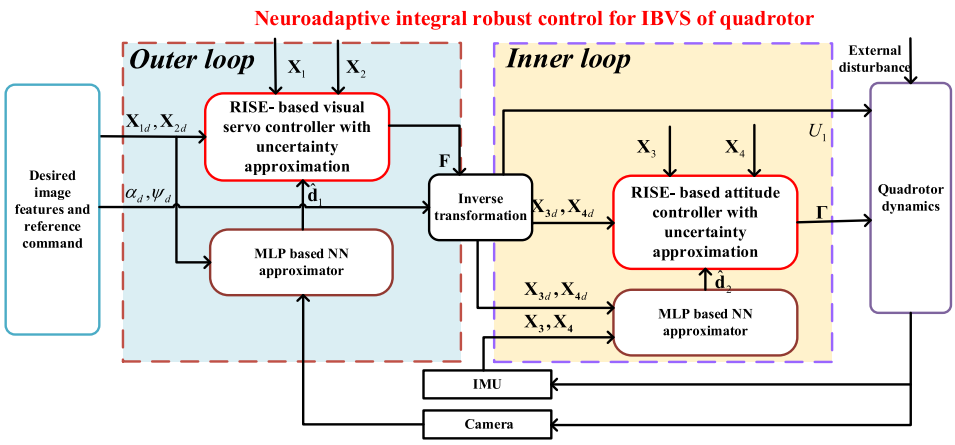

Fig. 2. The structure of neuroadaptive integral robust control for IBVS of quadrotor.

3.1. Outer-loop control

For the servo loop, let $\mathbf{q_d}$ be the desired image feature. And a neuroadaptive integral robust controller is designed with the purpose of regulating the real object feature $\mathbf{q}$ to match the desired feature $\mathbf{q_d}$. For convenience, we define $\mathbf{x_{1d}} = \mathbf{q_d}$. The tracking error and auxiliary filtered error for the outer loop are

$$\mathbf{e_1} = \mathbf{x_1} - \mathbf{x_{1d}}, \mathbf{e_2} = \dot{\mathbf{e}}_1 + k_{11}\mathbf{e_1} = \mathbf{x_2} - \bar{\mathbf{x}}_2, \bar{\mathbf{x}}_2 \triangleq \dot{\mathbf{x}}_{1d} - k_{11}\mathbf{e_1}, \mathbf{r_1} = \dot{\mathbf{e}}_2 + k_{12}\mathbf{e_2} \tag{10}$$

where k_{11}, k_{12} represent positive feedback arguments. The error signal $\mathbf{r_1}$ is constructed in (10) to acquire additional design freedom. Using Eq. (7) and (10), we can deduce the specific formulation of $\mathbf{r_1}$:

$$z^*\mathbf{r_1} = \mathbf{f_V} + \mathbf{F} + \mathbf{d_1} - z^*\dot{\bar{\mathbf{x}}}_2 + k_{12}z^*\mathbf{e_2} \tag{11}$$

For (11), the unknown lumped uncertainties $\mathbf{d_1}$ is the main difficulty of IBVS control design. In the traditional RISE design, a high feedback gain is generally required to cut down the effect of $\mathbf{d_1}$, which is not desired in practice.

Here an MLP-NN is first introduced to identify the lumped uncertainties. With the RBFNN (8), treating the image feature $\mathbf{X}_1$ and its optic flow $\mathbf{X}_2$ as the inputs, we can rewrite the form of $\mathbf{d_1}$ for each channel of outer loop as below:

$$\mathbf{d_1}(i) = \mathbf{W}_{oi}^{*T}\mathbf{h}_{oi}(\mathbf{X_1}(i), \mathbf{X_2}(i)) + \varsigma_{1i}, |\varsigma_{1i}| \leqslant \varsigma_{1M}, i = 1,2,3 \tag{12}$$

where $\mathbf{d_1}(i), i = 1,2,3$ denotes the lumped uncertainties for X_i, Y_i and Z_i directions respectively. $\mathbf{X_1}(i), \mathbf{X_2}(i)$ are the inputs of NN for each channel. $\mathbf{W}_{oi}^* = \left[w_{oi1}, w_{oi2}, \ldots w_{oip}\right]^T \in \mathbb{R}^p, i = 1,2,3$ is the desired weight vector for each channel of outer loop respectively. $\mathbf{h}_{oi} = \left[h_{oi1}(\mathbf{X_1}(i), \mathbf{X_2}(i)), h_{oi2}(\mathbf{X_1}(i), \mathbf{X_2}(i)), \ldots h_{oip}(\mathbf{X_1}(i), \mathbf{X_2}(i))\right]^T \in \mathbb{R}^p, i = 1,2,3,$ with $h_{oij}, j = 1,2,\ldots p$ possessing the same formula as (9). ς_{1i} expresses the approximation error for each channel and ς_{1M} represents the upper bound of error.

Based on the discussion on the MPL technology in Section 2.2, we define $\boldsymbol{\varphi}_1 = [\varphi_{11}, \varphi_{12}, \varphi_{13}]^T \in \mathbb{R}^{3\times 1}$ to replace the weight vector $\mathbf{W}_o^*$. And an adaptive law is designed for timely estimating the $\boldsymbol{\varphi}_1$ with the following formulation:

$$\dot{\hat{\boldsymbol{\varphi}}}_{1i} = \frac{\lambda_1}{2}\mathbf{e}_1(i)\mathbf{h}_{oi}^T(\mathbf{X}_1(i), \mathbf{X}_2(i))\mathbf{h}_{oi}(\mathbf{X}_1(i), \mathbf{X}_2(i)) - 2\hat{\boldsymbol{\varphi}}_{1i}, i = 1,2,3 \tag{13}$$

where $\hat{\boldsymbol{\varphi}}_{1i}$ is the estimated value of $\boldsymbol{\varphi_1}$ for each subsystem, and λ_1 denotes an adaptive gain. The estimation error of $\boldsymbol{\varphi}_1$ can be given as

$$\tilde{\boldsymbol{\varphi}}_1 = \hat{\boldsymbol{\varphi}}_1 - \boldsymbol{\varphi}_1 \tag{14}$$

With the adaptive NN weights in (13), the estimation of the lumped uncertainties $\mathbf{d_1}$ for each loop can be obtained:

$$\hat{\mathbf{d}}_1(\mathrm{i}) = \frac{1}{2}\hat{\boldsymbol{\varphi}}_{1i}\mathbf{h}_{oi}^T(\mathbf{X}_1(\mathrm{i}), \mathbf{X}_2(\mathrm{i}))\mathbf{h}_{oi}(\mathbf{X}_1(\mathrm{i}), \mathbf{X}_2(\mathrm{i})), i = 1,2,3 \tag{15}$$

Remark 5. In contrast to the classical NN approximation [30,31] that requires many nodes and lots of tuning parameters, the designed MLP-NN preserves lower computational loads and better practicality. As seen from the neural approximation (15) and the adaptive updating law (13), with the MLP technology, only one tuning parameter is required for the outer loop.

Then an integrated RISE controller with MLP-NN is constructed as follows:

$$\mathbf{F} = \mathbf{F}_a + \mathbf{F}_s, \mathbf{F}_a = -\mathbf{f_V} + z^*\dot{\bar{\mathbf{x}}}_2 - \hat{\mathbf{d}}_1\mathbf{F}_s = \mathbf{F}_{s1} + \mathbf{F}_{s2}, \mathbf{F}_{s1} = -(k_{r1} + k_{12}z^*)\mathbf{e}_2 \tag{16}$$

where k_{r1} is a positive gain. In (16), $\mathbf{F}_a$ represents a model based feedforward control law to counteract the model nonlinearity and lumped uncertainties, and $\mathbf{F}_s$ functions as a robust feedback control law, where $\mathbf{F}_{s1}$ is denoted as a traditional linear feedback term that ensures the unperturbed vision-quadrotor dynamics stabilization and $\mathbf{F}_{s2}$ is a robustified nonlinear control term utilized to restrain the residual approximation error:

$$\mathbf{F}_{s2} = -\int_0^t k_{r1}k_{12}\mathbf{e_2} + \beta_1 sign(\mathbf{e_2})d\zeta \tag{17}$$

where $\beta_1 > 0$ represents a positive feedback parameter, and $sign(\mathbf{e_2})$ is a nonzero signum function with respect to $\mathbf{e_2}$.

With the RISE neural control law (16) for outer-loop, (11) can be rewritten as:

$$\begin{aligned} z^*\mathbf{r_1} &= \mathbf{f_V} + \mathbf{F} + \mathbf{d_1} - z^*\dot{\bar{\mathbf{x}}}_2 + k_{12}z^*\mathbf{e_2} = \mathbf{f_V} + \mathbf{F}_a + \mathbf{F}_s - z^*\dot{\bar{\mathbf{x}}}_2 + k_{12}z^*\mathbf{e_2} = \mathbf{d_1} - \hat{\mathbf{d}}_1 - k_{r1}\mathbf{e}_2 + \mathbf{F}_{s2} \\ &= -k_{r1}\mathbf{e}_2 - \int_0^t k_{r1}k_{12}\mathbf{e_2} + \beta_1 sign(\mathbf{e_2})d\zeta + \boldsymbol{\varsigma}_1 \end{aligned} \tag{18}$$

Then the derivative dynamics of $\mathbf{r}_1$ can be given by

$$z^*\dot{\mathbf{r}}_1 = -k_{r1}\mathbf{r}_1 - \beta_1 sign(\mathbf{e}_2) + \dot{\boldsymbol{\varsigma}}_1 \tag{19}$$

As seen from (19) that if the estimation error ς_{1i} equals to 0, the nonlinear feedback term $\mathbf{F}_{s2}$ is needless, and the linear feedback term is enough to ensure the stability of visual quadrotor system. Nevertheless, it is inferred from analysis of NN approximation in Section 2.2 that the approximation error ς_{1i} is bounded and its upper bound is limited by ς_{1M}. Thus, the robust nonlinear control term is essential to overcome the residual approximation error for enhancing the robustness of system. By doing so, the design conservatism on selecting the integral robust feedback gain can also be relaxed, which will be revealed from the following assumption.

Assumption 2 (*[26,27]*). Based on the NN analysis and assumption 1, we give the assumption on the lumped approximation error ς_{1i}:

$$|\dot{\varsigma}_{1i}| \leqslant \delta_1, \quad |\ddot{\varsigma}_{1i}| \leqslant \delta_2, i = x, y, z \tag{20}$$

where δ_1, δ_2 are some positive constants.

Remark 6. For the traditional RISE design investigated in [26–28], the feedforward compensation term $\mathbf{F}_a$ does not contain the uncertainty approximation $\hat{\mathbf{d}}_1$, and the uncertainty term is primarily addressed via the robust feedback term $\mathbf{F}_s$. Consequently, the robustness of the controller is still limited when facing vast range of uncertainties. By contrast, in the designed neuroadaptive integral robust control, by means of the approximation capability of NN, uncertainty estimation is incorporated into the RISE control law via a feedforward manner. Using this methodology, the conservative assumptions on RISE in the existing literatures [26] that the feedback gain of RISE should be larger than the upper bound for time-derivatives of lumped uncertainties is relaxed.

Practically, the virtual control vector $\mathbf{F}$ are related to the desired force vector. Its magnitude is the applied thrust force U_1, and its orientation relies on the desired body attitude $(\phi_d, \theta_d, \psi_d)$, with ϕ_d, θ_d and ψ_d respectively being the reference roll, pitch and yaw signals.

Recalling the Eq. (4), the elements of $\mathbf{F} = [F_1, F_2, F_3]^T$ can be written as:

$$\begin{cases} F_1 = -\frac{U_1}{m}(c(\psi_d)s(\theta_d)c(\phi_d) + s(\psi_d)s(\phi_d)) \\ F_2 = -\frac{U_1}{m}(s(\psi_d)s(\theta_d)c(\phi_d) - c(\psi_d)s(\phi_d)) \\ F_3 = g - \frac{U_1}{m}(c(\theta_d)c(\phi_d)) \end{cases} \tag{21}$$

In consequence, the thrust force U_1 and the desired attitudes ϕ_d, θ_d can be deduced via inverse transformation of (21) as follows:

$$\begin{cases} U_1 = m\sqrt{F_1^2 + F_2^2 + F_3^2} \\ \phi_d = -\arcsin\left[\frac{m}{U_1}(F_1 s(\psi_d) - F_2 c(\psi_d))\right] \\ \theta_d = -\arctan\left[\frac{1}{g - F_3}(F_1 c(\psi_d) + F_2 s(\psi_d))\right] \end{cases} \tag{22}$$

where the yaw command ψ_d is provided by some high-level guidance system.

3.2. Inner-loop control

Considering the visual quadrotor dynamics in (7), here an attitude regulation law is designed for tracking the desired attitude commands, where the reference roll and pitch signals are generated from the IBVS controller of outer loop, while yaw motion is controlled via the defined image feature in (3).

Define the reference signals for rotational dynamics as $\bar{\mathbf{x}}_3 = [\phi_d, \theta_d, \alpha_d]^T$, the tracking error and auxiliary filtered error for the inner loop can be given by:

$$\mathbf{e}_3 = \mathbf{x}_3 - \bar{\mathbf{x}}_3, \mathbf{e}_4 = \mathbf{p}^{-1}(\dot{\mathbf{e}}_3 + k_{21}\mathbf{e}_3) = \mathbf{x}_4 - \bar{\mathbf{x}}_4, \bar{\mathbf{x}}_4 \triangleq \mathbf{p}^{-1}\left(\dot{\bar{\mathbf{x}}}_3 - k_{21}\mathbf{e}_3\right), \mathbf{r}_2 = \dot{\mathbf{e}}_4 + k_{22}\mathbf{e}_4 \tag{23}$$

where k_{21}, k_{22} are positive feedback gains. It is worth pointing out that an auxiliary error signal $\mathbf{r}_2$ is introduced in (23) to receive additional design freedom. Notably, $\mathbf{r}_2$ will be crucial for the subsequent stability analysis of RISE despite it does not appear in the RISE control design. Then we can deduce the expansion formulation of $\mathbf{r}_2$ using Eqs. (7) and (23):

$$\mathbf{r}_2 = \mathbf{f}_\gamma + \bar{\mathbf{\Gamma}} + \mathbf{d}_2 - \dot{\bar{\mathbf{x}}}_4 + k_{22}\mathbf{e}_4 \tag{24}$$

Following the same line with the NN approximator deigned for outer loop in (15), we give the uncertainty estimation for each channel of attitude loop based on MLP:

$$\hat{\mathbf{d}}_2(i) = \frac{1}{2}\hat{\boldsymbol{\varphi}}_{2i}\mathbf{h}_{li}^T(\mathbf{X}_3(i), \mathbf{X}_4(i))\mathbf{h}_{li}(\mathbf{X}_3(i), \mathbf{X}_4(i)), i = 1, 2, 3 \tag{25}$$

where $\hat{\mathbf{d}}_2(i)$, $i = 1,2,3$ means the estimated value of lumped uncertainties for ϕ, β and α channels respectively. $\hat{\boldsymbol{\varphi}}_{2i}$ denotes the adaptive estimation of NN weight $\boldsymbol{\varphi}_{2i}$ for each channel, $\mathbf{h}_{li} = \left[h_{li1}(\mathbf{X}_3(i),\mathbf{X}_4(i)), h_{li2}(\mathbf{X}_3(i),\mathbf{X}_4(i)), \ldots h_{lip}(\mathbf{X}_3(i),\mathbf{X}_4(i))\right]^T \in \mathbb{R}^p$, $i = 1,2,3$, with h_{lij}, $j = 1,2,\ldots p$ enjoying the same expression with (9). $\mathbf{X}_3(i)$, $\mathbf{X}_4(i)$, $i = 1,2,3$ represent the inputs of NN estimator in each channel. Notably, there still exists an approximation error between the uncertainty estimation $\hat{\mathbf{d}}_2$ and its actual value $\mathbf{d}_2$, which can be denoted as ς_{2i} for each subsystem with $|\boldsymbol{\varsigma}_{2i}| \leqslant \varsigma_{2M}$.

The adaptive law for updating the weights $\hat{\boldsymbol{\varphi}}_2$ in each channel is

$$\dot{\hat{\boldsymbol{\varphi}}}_{2i} = \frac{\lambda_2}{2}\mathbf{e}_3(i)\mathbf{h}_{li}^T(\mathbf{X}_3(i),\mathbf{X}_4(i))\mathbf{h}_{li}(\mathbf{X}_3(i),\mathbf{X}_4(i)) - 2\hat{\boldsymbol{\varphi}}_{2i}, i = 1,2,3 \tag{26}$$

with $\hat{\boldsymbol{\varphi}}_{2i}, i = 1,2,3$ being the estimated values of the weights $\boldsymbol{\varphi}_2$ in each subsystem, and λ_2 denotes a design parameter. Define the estimation error of $\boldsymbol{\varphi}_2$ as

$$\tilde{\boldsymbol{\varphi}}_2 = \hat{\boldsymbol{\varphi}}_2 - \boldsymbol{\varphi}_2 \tag{27}$$

Then the RISE controller with NN approximation can be given as

$$\begin{cases} \bar{\boldsymbol{\Gamma}} = \bar{\boldsymbol{\Gamma}}_a + \bar{\boldsymbol{\Gamma}}_s \\ \bar{\boldsymbol{\Gamma}}_a = -\mathbf{f}_\gamma + \dot{\bar{\mathbf{x}}}_4 - \hat{\mathbf{d}}_2 \\ \bar{\boldsymbol{\Gamma}}_s = \bar{\boldsymbol{\Gamma}}_{s1} + \bar{\boldsymbol{\Gamma}}_{s2} \\ \bar{\boldsymbol{\Gamma}}_{s1} = -(k_{r2} + k_{21})\mathbf{e}_4 \\ \bar{\boldsymbol{\Gamma}}_{s2} = -\int_0^t k_{r2}k_{22}\mathbf{e}_4 + \beta_2 sign(\mathbf{e}_4)d\zeta \end{cases} \tag{28}$$

where k_{r2}, β_2 are positive feedback gains.

Following the same line with the RISE control design for the outer loop, $\bar{\boldsymbol{\Gamma}}_a$ represents a model based feedforward term to eliminate the nonlinear behavior and compensate for the uncertainty in the inner loop, $\bar{\boldsymbol{\Gamma}}_s$ denotes a robust feedback control law, where $\bar{\boldsymbol{\Gamma}}_{s1}$ is a traditional linear feedback control term that stabilizes the vision-quadrotor dynamics, and $\bar{\boldsymbol{\Gamma}}_{s2}$ functions as a robustified nonlinear control law utilized to suppress the residual estimation errors. Then the rotational torque control vector $\boldsymbol{\Gamma}$ can be calculated from the virtual control vector $\bar{\boldsymbol{\Gamma}}$ as follows:

$$\boldsymbol{\Gamma} = \mathbf{J}\bar{\mathbf{g}}_2^{-1}\bar{\boldsymbol{\Gamma}} \tag{29}$$

Using the RISE neural control in (28), the expression of $\mathbf{r_2}$ can be rewritten as

$$\mathbf{r_2} = \mathbf{f}_\gamma + \bar{\boldsymbol{\Gamma}} + \mathbf{d_2} - \dot{\bar{\mathbf{x}}}_4 + k_{22}\mathbf{e_4} = \mathbf{d}_2 - \hat{\mathbf{d}}_2 - k_{r2}\mathbf{e}_4 + \bar{\boldsymbol{\Gamma}}_{s2} = -k_{r2}\mathbf{e}_4 - \int_0^t k_{r2}k_{22}\mathbf{e}_4 + \beta_2 sign(\mathbf{e}_4)d\zeta + \boldsymbol{\varsigma}_2 \tag{30}$$

And its time-derivative is

$$\dot{\mathbf{r}}_2 = -k_{r2}\mathbf{r}_2 - \beta_2 sign(\mathbf{e}_4) + \dot{\boldsymbol{\varsigma}}_2 \tag{31}$$

Similar with the assumption 2 for outer-loop, we have the following assumption for the rotational loop as well:

Assumption 3. Consider the assumption 1 and the analysis of NN, the lumped approximation error ς_2 is assumed as smooth enough such that

$$|\dot{\varsigma}_{2i}| \leqslant \delta_3, |\ddot{\varsigma}_{2i}| \leqslant \delta_4, i = \phi, \theta, \alpha \tag{32}$$

where δ_3, δ_4 are some positive constants.

4. Stability analysis

The stability proof of the present neuroadaptive integral robust based IBVS controller is given in this section. Before that, we introduce the following lemma which will be invoked later.

Lemma 2. Considering the servo and rotational loop, define the auxiliary functions $\mathbf{L}_1(t), \mathbf{L}_2(t)$ as

$$\mathbf{L}_1(t) = \mathbf{r}_1^T[\dot{\boldsymbol{\varsigma}}_1 - \beta_1 sign(\mathbf{e}_2)] + D_{M1}, \mathbf{L}_2(t) = \mathbf{r}_2^T[\dot{\boldsymbol{\varsigma}}_2 - \beta_2 sign(\mathbf{e}_4)] + D_{M2} \tag{33}$$

where $D_{M1} = \frac{\|\boldsymbol{\varphi}_1\|^2}{\lambda_1}$, $D_{M2} = \frac{\|\boldsymbol{\varphi}_2\|^2}{\lambda_2}$.

Choose the nonlinear integral signum feedback gains β_1, β_2 adequately large such that

$$\beta_1 \geqslant \delta_1 + \frac{1}{k_{12}}\delta_2, \beta_2 \geqslant \delta_3 + \frac{1}{k_{22}}\delta_4 \tag{34}$$

Then the following defined functions $\mathbf{P}_1(t), \mathbf{P}_2(t)$ will hold positive all the time.

$$\mathbf{P}_1(t) = \beta_1 \| \mathbf{e}_2(0) \| - \mathbf{e}_2(0)\dot{\boldsymbol{\varsigma}}_1(0) - \int_0^t \mathbf{L}_1(\zeta)d\zeta,\ \mathbf{P}_2(t) = \beta_2 \| \mathbf{e}_4(0) \| - \mathbf{e}_4(0)\dot{\boldsymbol{\varsigma}}_2(0) - \int_0^t \mathbf{L}_2(\zeta)d\zeta \tag{35}$$

The details can be referred to [26,38].

Theorem 1. Considering the deigned MLP-NN (15), (25) and the composite RISE control law (16), for the given image feature commands (28), if the gains β_1, β_2 satisfy the inequality (34), the approximator parameters λ_1, λ_2 and the controller gains $k_{11}, k_{12}, k_{21}, k_{22}, k_{r1}, k_{r2}$ are appropriately selected such that the following defined matrix Λ keeps positive definite, then the proposed neuroadaptive integral robust control scheme can guarantee the boundness of all signals in the closed-loop operation, and the asymptotic tracking performance can be achieved even in the presence of various uncertainties.

$$\Lambda = \begin{bmatrix} \Lambda_1 & 0 & 0 \\ 0 & \Lambda_2 & 0 \\ 0 & 0 & \Lambda_3 \end{bmatrix}, \Lambda_1 = \begin{bmatrix} k_{11} - \lambda_1 \hbar_o^4/4 & -0.5 & 0 \\ -0.5 & k_{12} & -0.5 \\ 0 & -0.5 & k_{r1} \end{bmatrix} \Lambda_2 = \begin{bmatrix} k_{21} - \lambda_2 \hbar_I^4/4 & -0.5 & 0 \\ -0.5 & k_{22} & -0.5 \\ 0 & -0.5 & k_{r2} \end{bmatrix}, \Lambda_3 = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & \frac{3}{2} \end{bmatrix}^T \tag{36}$$

Proof. First, define the following Lyapunov candidate function:

$$V = 0.5\mathbf{e}_1^T\mathbf{e}_1 + 0.5\mathbf{e}_2^T\mathbf{e}_2 + 0.5z^*\mathbf{r}_1^T\mathbf{r}_1 + \frac{\tilde{\boldsymbol{\varphi}}_1^T\tilde{\boldsymbol{\varphi}}_1}{2\lambda_1} + 0.5\mathbf{e}_3^T\mathbf{e}_3 + 0.5\mathbf{e}_4^T\mathbf{e}_4 + 0.5\mathbf{r}_2^T\mathbf{r}_2 + \frac{\tilde{\boldsymbol{\varphi}}_2^T\tilde{\boldsymbol{\varphi}}_2}{2\lambda_2} + \mathbf{P}_1 + \mathbf{P}_2 \tag{37}$$

By recalling Eq. (13), (16), (19), (26), (31), and (33), we can give the time-derivative of V as follows:

$$\begin{aligned} \dot{V} &= \mathbf{e}_1^T\dot{\mathbf{e}}_1 + \mathbf{e}_2^T\dot{\mathbf{e}}_2 + z^*\mathbf{r}_1^T\dot{\mathbf{r}}_1 + \frac{\tilde{\boldsymbol{\varphi}}_1^T\dot{\tilde{\boldsymbol{\varphi}}}_1^T}{\lambda_1} + \mathbf{e}_3^T\dot{\mathbf{e}}_3 + \mathbf{e}_4^T\dot{\mathbf{e}}_4 + \mathbf{r}_2^T\dot{\mathbf{r}}_2 + \dot{\mathbf{P}}_1 + \dot{\mathbf{P}}_2 + \frac{\tilde{\boldsymbol{\varphi}}_2^T\dot{\tilde{\boldsymbol{\varphi}}}_2^T}{\lambda_2} \\ &= \mathbf{e}_1^T(\mathbf{e}_2 - k_{11}\mathbf{e}_1) + \mathbf{e}_2^T(\mathbf{r}_1 - k_{12}\mathbf{e}_2) + \mathbf{r}_1^T(-k_{r1}\mathbf{r}_1 - \beta_1 sign(\mathbf{e}_2) + \dot{\boldsymbol{\varsigma}}_1) + \frac{\tilde{\boldsymbol{\varphi}}_1^T}{\lambda_1}\left(\frac{\lambda_1}{2}\mathbf{e}_1\mathbf{h}_o^T\mathbf{h}_o - 2\hat{\boldsymbol{\varphi}}_1\right) - \mathbf{L}_1\mathbf{e}_3^T(\mathbf{e}_4 - k_{21}\mathbf{e}_3) \\ &\quad + \mathbf{e}_4^T(\mathbf{r}_2 - k_{22}\mathbf{e}_4) + \mathbf{r}_2^T(-k_{r2}\mathbf{r}_2 - \beta_2 sign(\mathbf{e}_4) + \dot{\boldsymbol{\varsigma}}_2) + \frac{\tilde{\boldsymbol{\varphi}}_2^T}{\lambda_2}\left(\frac{\lambda_2}{2}\mathbf{e}_3\mathbf{h}_I^T\mathbf{h}_I - 2\hat{\boldsymbol{\varphi}}_2\right) - \mathbf{L}_2 \end{aligned} \tag{38}$$

Referring to Remark 2, we can conclude that there exist positive constants $\hbar_o, \hbar_I$ such that $\| \mathbf{h}_o \| \leqslant \hbar_o, \| \mathbf{h}_I \| \leqslant \hbar_I$. Using Young's inequality, one has

$$\frac{\tilde{\boldsymbol{\varphi}}_1^T}{2}\mathbf{e}_1\mathbf{h}_o^T\mathbf{h}_o \leqslant \frac{\hbar_o^2}{2}\tilde{\boldsymbol{\varphi}}_1^T\mathbf{e}_1 \leqslant \frac{2\lambda_1}{2}\left(\frac{\hbar_o^2}{2}\mathbf{e}_1\right)^2 + \frac{1}{2(2\lambda_1)}\| \tilde{\boldsymbol{\varphi}}_1 \|^2 = \frac{\lambda_1\hbar_o^4\| \mathbf{e}_1 \|^2}{4} + \frac{\| \tilde{\boldsymbol{\varphi}}_1 \|^2}{4\lambda_1} \tag{39}$$

$$\frac{\tilde{\boldsymbol{\varphi}}_2^T}{2}\mathbf{e}_3\mathbf{h}_I^T\mathbf{h}_I \leqslant \frac{\hbar_I^2}{2}\tilde{\boldsymbol{\varphi}}_2^T\mathbf{e}_3 \leqslant \frac{2\lambda_2}{2}\left(\frac{\hbar_I^2}{2}\mathbf{e}_3\right)^2 + \frac{1}{2(2\lambda_2)}\| \tilde{\boldsymbol{\varphi}}_2 \|^2 = \frac{\lambda_2\hbar_I^4\| \mathbf{e}_3 \|^2}{4} + \frac{\| \tilde{\boldsymbol{\varphi}}_2 \|^2}{4\lambda_2} \tag{40}$$

Based on (33), (39) and (40), the time-derivative of V can be derived as

$$\begin{aligned} \dot{V} &= \mathbf{e}_1^T(\mathbf{e}_2 - k_{11}\mathbf{e}_1) + \mathbf{e}_2^T(\mathbf{r}_1 - k_{12}\mathbf{e}_2) + \mathbf{r}_1^T(-k_{r1}\mathbf{r}_1 - \beta_1 sign(\mathbf{e}_2) + \dot{\boldsymbol{\varsigma}}_1) + \frac{\tilde{\boldsymbol{\varphi}}_1^T}{\lambda_1}\left(\frac{\lambda_1}{2}\mathbf{e}_1\mathbf{h}_o^T\mathbf{h}_o - 2\hat{\boldsymbol{\varphi}}_1\right) - \mathbf{r}_1^T[\dot{\boldsymbol{\varsigma}}_1 - \beta_1 sign(\mathbf{e}_2)] \\ &\quad - \frac{\| \boldsymbol{\varphi}_1 \|^2}{\lambda_1} + \mathbf{e}_3^T(\mathbf{e}_4 - k_{21}\mathbf{e}_3) + \mathbf{e}_4^T(\mathbf{r}_2 - k_{22}\mathbf{e}_4) + \mathbf{r}_2^T(-k_{r2}\mathbf{r}_2 - \beta_2 sign(\mathbf{e}_4) + \dot{\boldsymbol{\varsigma}}_2) + \frac{\tilde{\boldsymbol{\varphi}}_2^T}{\lambda_2}\left(\frac{\lambda_2}{2}\mathbf{e}_3\mathbf{h}_I^T\mathbf{h}_I - 2\hat{\boldsymbol{\varphi}}_2\right) \\ &\quad - \mathbf{r}_2^T[\dot{\boldsymbol{\varsigma}}_2 - \beta_2 sign(\mathbf{e}_4)] - \frac{\| \boldsymbol{\varphi}_2 \|^2}{\lambda_2} \leqslant -k_{11}\| \mathbf{e}_1 \|^2 - k_{12}\| \mathbf{e}_2 \|^2 - k_{r1}\| \mathbf{r}_1 \|^2 - k_{21}\| \mathbf{e}_3 \|^2 - k_{22}\| \mathbf{e}_4 \|^2 - k_{r2}\| \mathbf{r}_2 \|^2 \\ &\quad + \mathbf{e}_1^T\mathbf{e}_2 + \mathbf{e}_2^T\mathbf{r}_1 + \mathbf{e}_3^T\mathbf{e}_4 + \mathbf{e}_4^T\mathbf{r}_2 + \frac{\lambda_1\hbar_o^4\| \mathbf{e}_1 \|^2}{4} + \frac{\| \tilde{\boldsymbol{\varphi}}_1 \|^2}{4\lambda_1} - \frac{\| \tilde{\boldsymbol{\varphi}}_1 \|^2}{\lambda_1} + \frac{\| \boldsymbol{\varphi}_1 \|^2}{\lambda_1} - \frac{\| \boldsymbol{\varphi}_1 \|^2}{\lambda_1} + \frac{\lambda_2\hbar_I^4\| \mathbf{e}_3 \|^2}{4} + \frac{\| \tilde{\boldsymbol{\varphi}}_2 \|^2}{4\lambda_2} \\ &\quad - \frac{\| \tilde{\boldsymbol{\varphi}}_2 \|^2}{\lambda_2} + \frac{\| \boldsymbol{\varphi}_2 \|^2}{\lambda_2} - \frac{\| \boldsymbol{\varphi}_2 \|^2}{\lambda_2} = -\left(k_{11} - \frac{\lambda_1\hbar_o^4}{4}\right)\| \mathbf{e}_1 \|^2 - k_{12}\| \mathbf{e}_2 \|^2 - k_{r1}\| \mathbf{r}_1 \|^2 - \left(k_{21} - \frac{\lambda_2\hbar_I^4}{4}\right)\| \mathbf{e}_3 \|^2 \\ &\quad - k_{22}\| \mathbf{e}_4 \|^2 - k_{r2}\| \mathbf{r}_2 \|^2 + \mathbf{e}_1^T\mathbf{e}_2 + \mathbf{e}_2^T\mathbf{r}_1 + \mathbf{e}_3^T\mathbf{e}_4 + \mathbf{e}_4^T\mathbf{r}_2 - \frac{3}{4\lambda_1}\| \tilde{\boldsymbol{\varphi}}_1 \|^2 - \frac{3}{4\lambda_2}\| \tilde{\boldsymbol{\varphi}}_2 \|^2 \end{aligned} \tag{41}$$

By means of the positive definite matrix Λ defined in (40), it can be derived that

$$\dot{V} \leqslant -\chi^T \Lambda \chi \leqslant -\lambda_{\min}(\Lambda)\left(\| \mathbf{e}_1 \|^2 + \| \mathbf{e}_2 \|^2 + \| \mathbf{r}_1 \|^2 + \| \mathbf{e}_3 \|^2 + \| \mathbf{e}_4 \|^2 + \| \mathbf{r}_2 \|^2 + \frac{\| \tilde{\boldsymbol{\varphi}}_1 \|^2}{2\lambda_1} + \frac{\| \tilde{\boldsymbol{\varphi}}_2 \|^2}{2\lambda_2} \right) \triangleq W \tag{42}$$

where $\chi = [\| \mathbf{e}_1 \|, \| \mathbf{e}_2 \|, \| \mathbf{r}_1 \|, \| \mathbf{e}_3 \|, \| \mathbf{e}_4 \|, \| \mathbf{r}_2 \|, \| \hat{\boldsymbol{\varphi}}_1 \|/2\lambda_1, \| \hat{\boldsymbol{\varphi}}_2 \|/2\lambda_2]^T$. $\lambda_{\min}(\cdot)$ means the minimum eigenvalue of a matrix.

Hence $V \in L_\infty$ and $W \in L_2$, leading to the result that the error χ is bounded, which also implies $W \in L_\infty$. Then, with the help of tracking error dynamics, we can conclude that the time-derivative of W is bounded, thus it can be further implied that W is consistently continuous. Using Barbalat's lemma [39], $W \to 0$ as $t \to 0$, and we can reach the result of Theorem 1.

This concludes the proof.

Remark 7. Unlike the existing researches [32,33,35] on the NN control that only achieve UUB, the proposed neuroadaptive integral robust control can guarantee the asymptotic stability of closed-loop system, with bounded and continuous control signals. This satisfactory property can lead to a more precise servo tracking performance.

5. Simulation results

The test on the proposed neuroadaptive integral robust based visual object tracking control is performed in MATLAB/SIMILINK environment. To simulate the actual operations as much as possible, the solver options are chosen as a fixed-step type with the size 0.02 s. As well, the solver is selected as a fourth order Runge–Kutta algorithm. Table 1 lists the main physical arguments of the quadrotor and camera. Using the onboard camera, the quadrotor is driven to perform a visual tracking task for the ground moving target. The visual target is considered as a rectangle consisting of four points, whose initial positions are $(-0.25, -0.2, 0)$m, $(-0.25, 0.2, 0)$m, $(0.25, -0.2, 0)$m, $(0.25, 0.2, 0)$m relative to the inertia frame. And the quadrotor is assumed to hover in a stably suspended mode initially and stay close to the object such that the observed object is inside the field of view (FOV) of its camera.

To realize a stable tracking for the moving target, we select the controller gains as $k_{11} = 2, k_{12} = 8, k_{r1} = 2, \beta_1 = 0.1$, $k_{21} = 30, k_{22} = 20, k_{r2} = 5, \beta_2 = 0.05$, the NN node numbers as $p = 100$, the center vectors for each loop as $\mathbf{c}_1 = [2, 2]^T$, the width as $b_j = 10$ and the adaptive parameters as $\lambda_1 = 0.02, \lambda_2 = 2$. The servo tracking performance of the proposed IBVS controller is validated in a noisy environment, where a zero-mean Gaussian white noise signal with variance of 0.0002 rad is added to the attitude control loop. Moreover, the considered parameterized uncertainties and external disturbances are given as

$\Delta k_x = 0.2k_x, \Delta k_y = 0.2k_y, \Delta k_z = 0.2k_z, \Delta k_\phi = 0.2k_\phi, \Delta k_\theta = 0.2k_\theta, \Delta k_\psi = 0.2k_\psi, \Delta l = 0.1l, \Delta c = 0.1c, d_{vx} = 0.5\sin(t), d_{vy} = 0.5\cos(t)$, $d_{vz} = 0.5\sin(t)\cos(t), d_{\Omega\phi} = 0.3\sin(0.5t), d_{\Omega\theta} = 0.3\cos(0.5t), d_{\Omega\psi} = 0.3\cos(0.5t)\sin(0.5t)$To validate the precise servo tracking performance and stability of the proposed neuroadaptive integral robust control, two set of experiments are performed:

- In the first set, the quadrotor is forced to track a ground object with complex motion trajectory for validating the effectiveness and flexibility of the designed neuroadaptive integral robust algorithm.
- In the second set, the simulation results are compared with traditional RISE control and the NN control to confirm the superiority of the designed controller in exhibiting a high performance servo tracking for the moving target that performs a circular maneuvering.

Table 1
Physical arguments of the quadrotor and camera.

Variable	Description	Value
m	Mass of the quadrotor	2kg
g	Gravity acceleration	9.8m/s^2
$\mathbf{\Pi}_1, \mathbf{\Pi}_2$	Damping matrices	$\mathbf{\Pi}_1 = diag(k_x, k_y, k_z) = diag(0.01, 0.01, 0.01)$Nms^2 $\mathbf{\Pi}_2 = diag(k_\phi, k_\theta, k_\psi) = diag(0.012, 0.012, 0.012)$Nms^2
$\mathbf{J}$	Inertial matrix	$\mathbf{J} = diag(J_1, J_2, J_3) = diag(0.16, 0.16, 0.32)$kgm^2
l	Distance from each rotor to the center of mass	0.4m
c	Force-to-moment factor	0.05
f	Focal length of camera	0.008m
u_N, n_N	Pixels number of camera	$(1280, 1024)$
o	Principal point of camera	$(640, 512)$

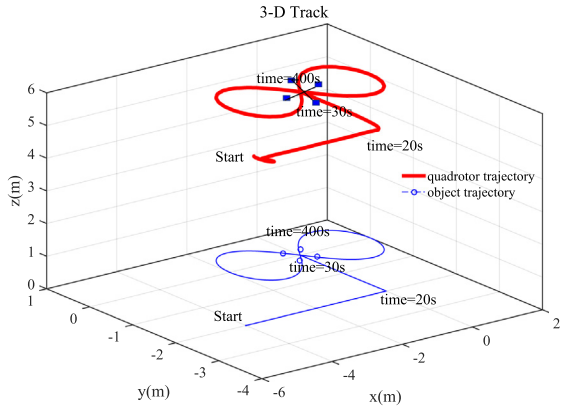

Fig. 3. Tracking trajectory of the quadrotor in 3D space.

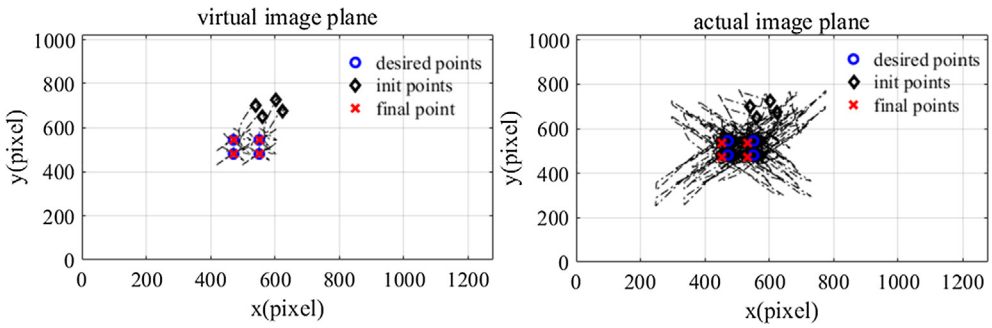

Fig. 4. Trajectories of feature points for the sinusoid path: (a) virtual image plane, (b) real image plane.

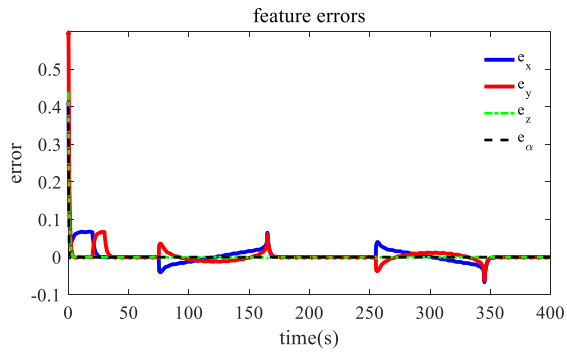

Fig. 5. Image matching errors.

5.1. Experiment 1

In this set of simulation, a complex targeting route involving straight lines and curves is designed to confirm the effectiveness and flexibility of the present controller for quadrotor. The position of the centroid of the four object points is time-varying as

$$\begin{cases} p_x = -4 + 0.2t, p_y = -2 \quad 0 < t \leqslant 20 \\ p_x = 0, p_y = -2 + 0.2(t-20) \quad 20 < t \leqslant 30 \\ p_x = 2\sqrt{\cos(4pi/360(t\text{ - }30) + pi}cos(2pi/360(t\text{ - }30) + pi/2), \\ p_y = 2\sqrt{\cos(4pi/360(t\text{ - }30) + pi}sin(2pi/360(t\text{ - }30) + pi/2) \quad 30 < t \leqslant 400 \end{cases} \tag{43}$$

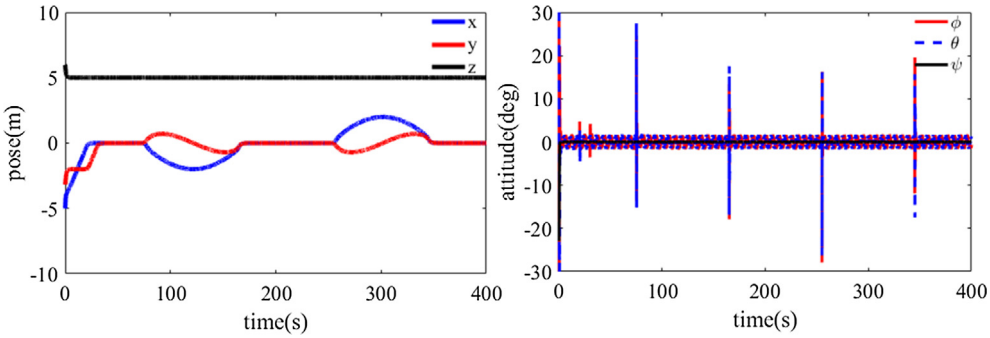

Fig. 6. Variation of position and attitude of quadrotor.

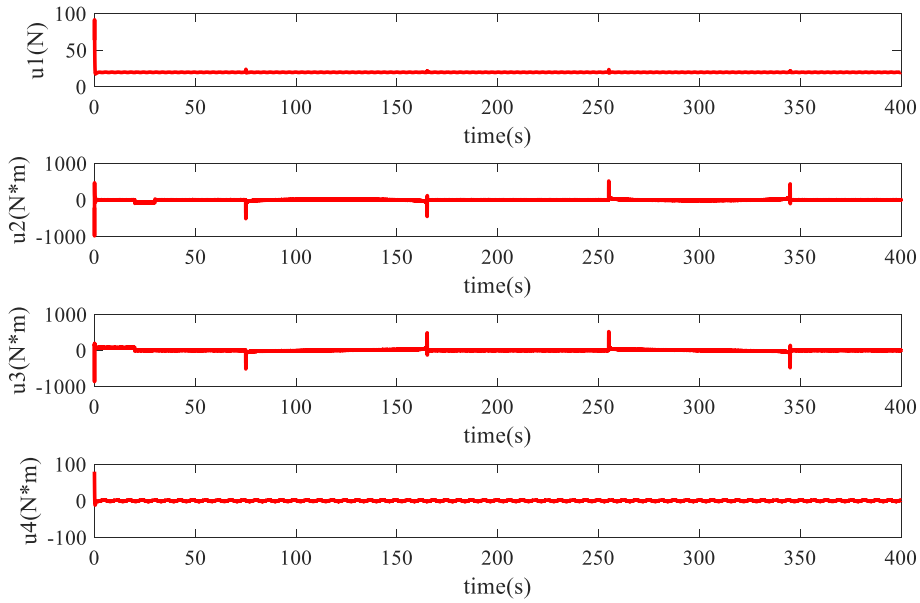

Fig. 7. Input signals of the quadrotor.

The initial position and attitude of quadrotor are set as $(-5.0, -3.0, -6.0)$m and $(0, 0, -0.4)$rad respectively in order to keep the feature points around the center of image plane. The image feature commands are set as $a^* = -4.5 \times 10^{-14}$, $[q_{xd}, q_{yd}, q_{zd}, \alpha_d]^T = [0, 0, 1, 0]^T$.The simulation results are displayed in Figs. 3–7.

As seen from Fig. 3, even facing the effect of strong external disturbances, parametric uncertainties and noises, the quadrotor still can provide a stable servo tracking for the ground moving target with complex targeting route involving straight lines and curves. This verifies the effectiveness and flexibility of the proposed controller by a fusion of NN approximation and RISE disturbance suppression. As well, the motions of feature points in the virtual and real image plane are revealed in Fig. 4. It can be seen that the feature points reach the desired points as expected, and all the features stay in the camera's field of view during the process of target tracking, demonstrating the precise image matching ability and strong robustness. These satisfactory properties can also be observed from the variation of image matching errors in Fig. 5, where all the errors converge to zero with asymptotic tracking performance. Notably, a drastic error change occurs around $time = 30s, 100s, 300s$, caused by the abrupt changes of object direction. Moreover, smoother point trajectories can be realized in the virtual image plane (Fig. 4(a)) in contrast to those in the real image plane (Fig. 4(b)) due to the under-actuated dynamics. For the virtual plane, a virtual camera isolating the roll and pitch angles is introduced to reproject the object points, hence the image motions will not be affected by the uncontrolled attitude motions. Furthermore, the satisfactory quadrotor position and attitude responses are revealed in Fig. 6. And the control inputs are displayed in Fig. 7. As seen, the designed controller can provide stable input signals for quadrotor, in spite of violent overshoots around $time = 30s, 100s, 300s$ raised by the sharp changes of object direction. The estimated weights of NN approximators in the outer and inner loops are given in Fig. 8, where the boundedness of all estimated signals can be ensured.

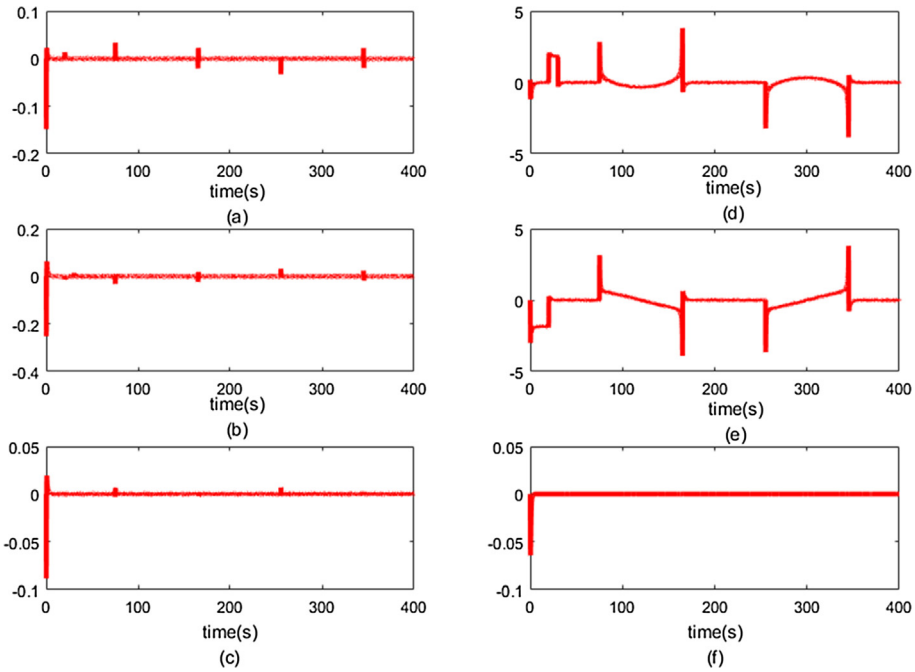

Fig. 8. The estimation of weights: (a) $\hat{\varphi}_{11}$, (b) $\hat{\varphi}_{12}$, (c) $\hat{\varphi}_{13}$, (d) $\hat{\varphi}_{21}$, (e) $\hat{\varphi}_{22}$, (f) $\hat{\varphi}_{23}$.

5.2. Experiment 2

To further validate the superiority of the designed controller in achieving precise servo tracking for the ground moving target, performance comparisons with traditional RISE control and NN control are carried out in the following.

Neuroadaptive integral robust: This is the designed IBVS controller in Section 3 by implanting the MLP-NN into the RISE control law. In which, the unknown lumped uncertainties are first estimated by the MLP-NN and compensated in the feed-forward loop, then the residual approximation errors are further restrained via the constructed RISE control law in feedback loop. The NN and RISE controller parameters are the same with those in Experiment 1.

RISE [30,38]: This is the traditional robust nonlinear control method without the uncertainty estimation, involving a model based feedforward control mechanism, a linear feedback term and a nonlinear integral signum feedback control law. The control gains for the RISE controller are consistent with the designed neuroadaptive integral robust control.

MLP-NN [34,37]: This is an adaptive NN control with a minimal learning parameter scheme for estimating the unknown uncertainties. Using the MLP technology, the NN approximation can circumvent the tedious weight estimation and the

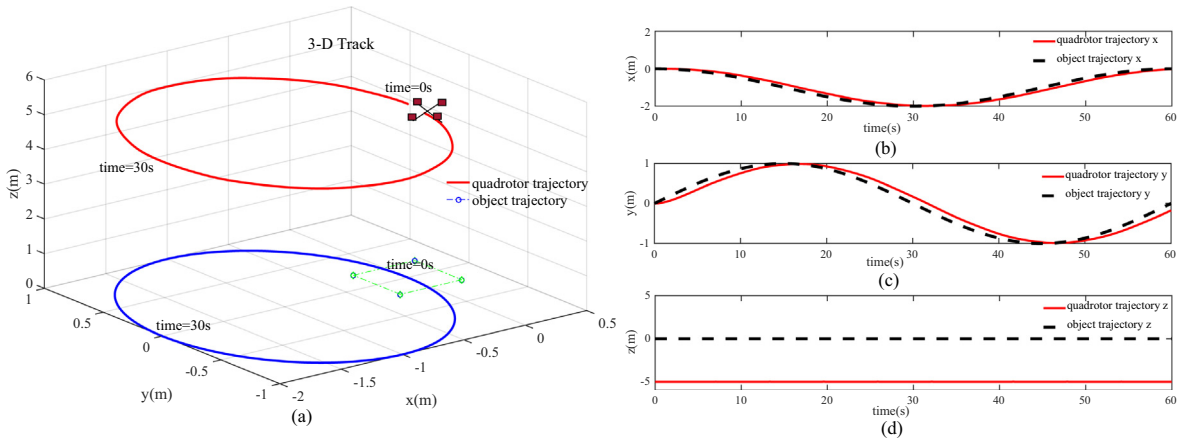

Fig. 9. Tracking trajectories of the quadrotor for the circular path using the neuroadaptive integral robust control: (a) 3D space, (b) X_i direction, (c) Y_i direction and (d) Z_i direction.

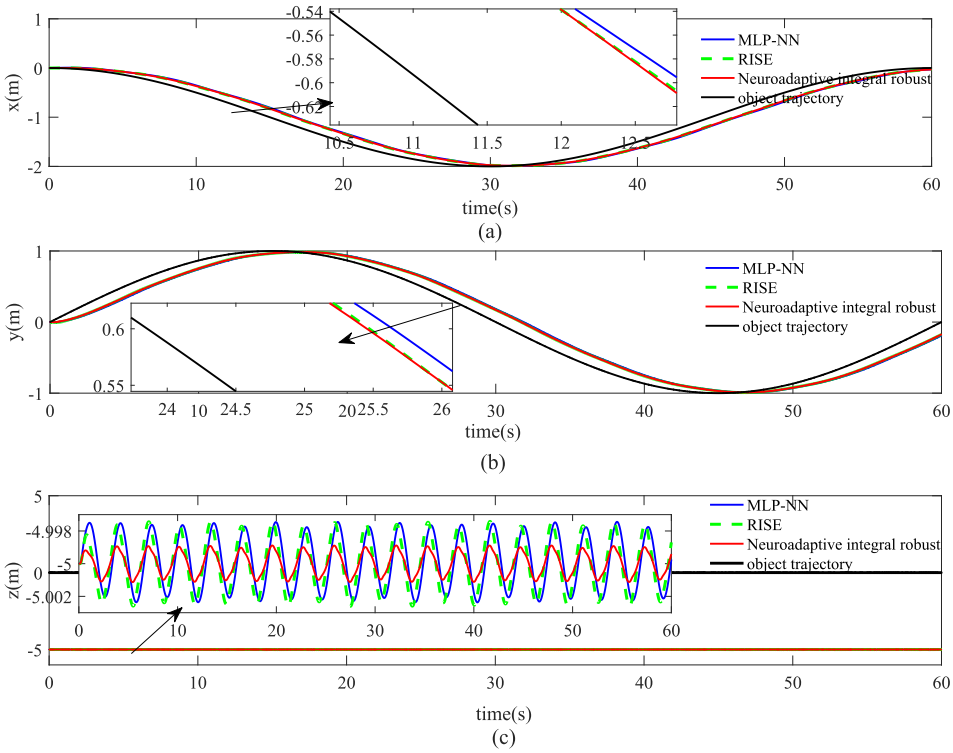

Fig. 10. Tracking trajectories of the quadrotor for the circular path using the considered three controller in: (a)X_i direction, (b) Y_i direction, (c)Z_i direction.

computational load is greatly reduced. For the sake of fairness, the NN parameters are the same with the neuroadaptive integral robust control.

The considered parameter uncertainties, external disturbances and noises are the same with Experiment 1. A ground target moving on a circle path with a radius of 3 m is selected to validate the visual tracking ability of quadrotor using the considered three controllers. The target moves with an angular rate of $\pi/30$rads^{-1}. The quadrotor is required to hold a stably suspended mode above the target at the beginning with initial position and attitude being $(0, 0, -5.0)$m and $(0, 0, 0)$rad respectively. The favorable image features are selected as $a^* = -4.5 \times 10^{-14}$ and $[q_{xd}, q_{yd}, q_{zd}, \alpha_d]^T = [0, 0, 1, 0]^T$.

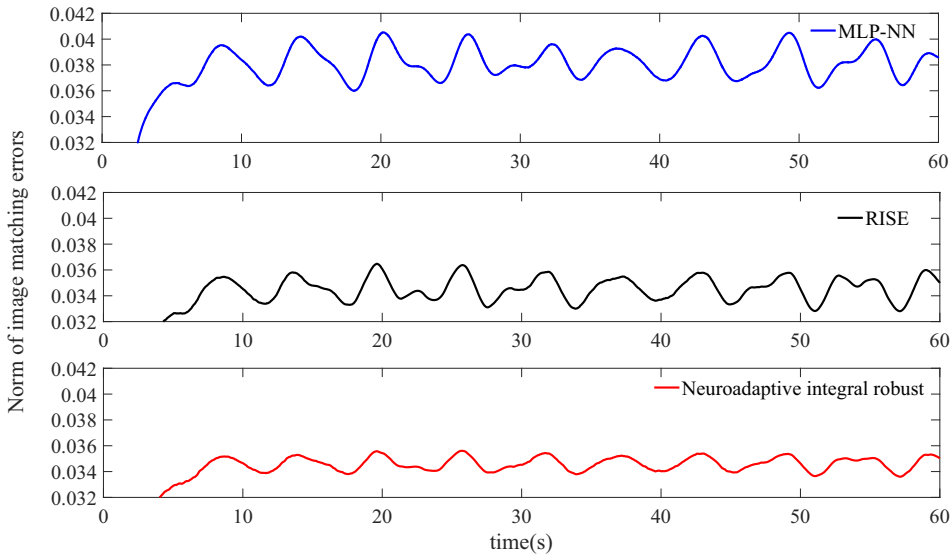

Fig. 11. Norm of image matching errors for considered three controllers.

Table 2
Standard deviation, mean and max of errors of image features in the IBVS system of quadrotors.

	Neuroadaptive integral robust	RISE	MLP-NN
STD	5.1708×10^{-4}	8.8445×10^{-4}	1.2×10^{-3}
MEAN	0.0340	0.0346	0.0382
MAX	0.0356	0.0365	0.0405

The 3D tracking trajectory of quadrotor for the ground target at the circular route using the proposed neuroadaptive integral robust is revealed in Fig. 9. It can be seen that the quadrotor with the present neuroadaptive integral robust based IBVS control has a stable 3D tracking capability for the ground moving target, with precise tracking results for the $X_i - Y_i - Z_i$ directions. In order to exhibit the superiority of the proposed method, the tracking results for $X_i - Y_i - Z_i$ directions using the considered three controllers are compared in Fig. 10. As observed, for the X_i and Y_i directions, the tracking trajectories with the neuroadaptive integral robust control are closer to the real object trajectories, comparing with the MLP-NN and RISE. For the Z_i direction, the quadrotor is required to hold at the presupposed constant height, i.e., $-5m$ for this paper. Obviously, the proposed neuroadaptive integral robust provides a more stable tracking ability in vertical direction with the smallest tracking error. To further illustrate the precise servo tracking performance and the strong robustness of proposed IBVS control, Fig. 11 illustrates the norm of errors of the image features with the considered three controllers. As seen, the proposed neuroadaptive integral robust controller exhibits the best servo performance with the smallest error. This satisfactory servo tracking performance can be attributed to the utilization of NN approximation in the RISE control law, where the unknown uncertainties are estimated and compensated in the feedforward loop and the residual estimation errors are further suppressed in the feedback loop. Hence the performance robustness with respect to parameterized uncertainties, external disturbances and noises can be enhanced. The performance indices during the stable tracking period for the three controllers are collected in Table 2, including the standard deviation, mean and max of normalized image error. It can be seen that the proposed neuroadaptive integral robust control outperforms the other two controllers, demonstrating that a remarkable enhancement in reducing the image matching errors is realized, which can further provide a stable servo tracking for the ground moving target with strong anti-disturbance ability.

6. Conclusion

In this paper, we consider the servo tracking control problem of quadrotors for the ground moving target in the presence of various uncertainties including parametric uncertainties, external disturbances and noises. By a fusion of NN approximation and RISE control, a neuroadaptive integral robust controller is proposed under the IBVS framework with asymptotic stability and strong robustness against uncertainties. Notably, by virtual of the designed uncertainty approximation, the RISE feedback gains are greatly reduced with a guaranteed high-precise tracking performance. Also, with the help of MLP technology, the number of tuned parameters in neural adaptive laws are decreased and the computational burden is relieved. The superiority of proposed control scheme is verified via designing abundant simulations and comparisons. Further works will focus on the IBVS control design for more complicated missions in reality, such as railway inspection, object searching, autonomously taking off and landing of quadrotor.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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